

10.6 ASYMPTOTIC STATES IN THE LSZ FORMULA

The derivation of the LSZ formula in LecNote4 treats $a_1^\dagger(\pm\infty)$ as creating one normalized particle, $a_1(\pm\infty)$ as annihilating the vacuum, and the vacuum as the same state at both ends. This note works out what the limit $t \rightarrow \pm\infty$ does and does not do. Two of the conditions on the field, $\langle 0|\varphi|0\rangle = 0$ and $\langle p|\varphi(0)|0\rangle = 1$, have to be imposed by hand, while the multi-particle condition takes care of itself; the reason is one formula, Eq. (10.6.2). We then estimate how fast the multi-particle part goes away, say what $\langle 0|0\rangle = 1$ means for the in and out vacua, and identify what the time-ordering step on LecNote4 p. 5 drops.

Notation follows LecNote4 and Srednicki §5: metric $(-, +, +, +)$,

$kx = \mathbf{k} \cdot \mathbf{x} - \omega t$, $\omega(\mathbf{k}) = \sqrt{\mathbf{k}^2 + m^2}$, $\widetilde{dk} = d^3k / ((2\pi)^3 2\omega)$, and

$$a_1^\dagger(t) = \int d^3k f_1(\mathbf{k}) a^\dagger(\mathbf{k}, t), \quad a^\dagger(\mathbf{k}, t) = -i \int d^3x e^{ikx} \overleftrightarrow{\partial}_0 \varphi(x), \quad (10.6.1)$$

with $f_1(\mathbf{k}) \propto \exp[-(\mathbf{k} - \mathbf{k}_1)^2 / 4\sigma^2]$, normalized so that in the free theory $\langle 0|a_1 a_1^\dagger|0\rangle = 1$.

ONE FORMULA FOR EVERY SECTOR

Let $|s\rangle$ be any eigenstate of P^μ , with momentum $\mathbf{p}_s$ and energy E_s .

Translation invariance gives $\langle s|\varphi(x)|0\rangle = e^{-ip_s x} \langle s|\varphi(0)|0\rangle$ (LecNote4 p. 7).

Since $e^{ikx} \overleftrightarrow{\partial}_0 e^{-ip_s x} = i(\omega + E_s) e^{i(k-p_s)x}$, the x integral in (10.6.1) gives $(2\pi)^3 \delta^3(\mathbf{k} - \mathbf{p}_s)$ and

$$\langle s|a_1^\dagger(t)|0\rangle = (2\pi)^3 [\omega(\mathbf{p}_s) + E_s] f_1(\mathbf{p}_s) \langle s|\varphi(0)|0\rangle e^{iK_s t}, \quad K_s \equiv E_s - \omega(\mathbf{p}_s). \quad (10.6.2)$$

Integrated against a probe packet $\psi_n^*(\mathbf{p})$, this is (5.23) of LecNote4. K_s depends only on the eigenvalues of P^μ , so (10.6.2) is a matrix element of the operator statement

$$a_1^\dagger(t)|0\rangle = e^{iKt} a_1^\dagger(0)|0\rangle, \quad K \equiv H - \sqrt{\mathbf{P}^2 + m^2}. \quad (10.6.3)$$

The time dependence of $a_1^\dagger(t)|0\rangle$ is generated by K , not by H . Two consequences follow.

First, K is Hermitian, so e^{iKt} is unitary and the norm of $a_1^\dagger(t)|0\rangle$ is the same at every t . Nothing is lost as $t \rightarrow \pm\infty$. The limit can only be a statement about overlaps with fixed states: a weak limit.

Second, for a fixed normalizable probe $|\psi\rangle$, the amplitude $\langle \psi|a_1^\dagger(t)|0\rangle$ is an average of e^{iKt} over the states in $|\psi\rangle$ and $a_1^\dagger(0)|0\rangle$. If K takes a continuum of values over those states, the phases cancel and the Riemann–Lebesgue

lemma (§10.5) sends the average to zero. If K takes a single value, the average is that one phase times a constant and does not decay. So the limit $t \rightarrow \pm\infty$ removes the continuous spectrum of K (strictly, its absolutely continuous part, which is all that smooth packets see) and nothing else.

SECTOR BY SECTOR

Vacuum. $E = 0$ and $\mathbf{p} = 0$, so $K = -m$. It is a single state, and there is nothing to average over. With $v \equiv \langle 0|\varphi(0)|0\rangle$, (10.6.2) gives

$$\langle 0|a_1^\dagger(t)|0\rangle = (2\pi)^3 m v f_1(\mathbf{0}) e^{-imt}. \quad (10.6.4)$$

The modulus is constant, and there is no limit unless $v f_1(\mathbf{0}) = 0$. For a generic packet this means $v = 0$, imposed by the shift $\varphi \rightarrow \varphi - v$ (LecNote4 p. 7).

One particle. $E = \omega(\mathbf{p})$ exactly, because $a^\dagger$ was built with the same m . So $K = 0$ for every momentum on the mass shell: the one-particle states form a continuum of H but a single point, $K = 0$, of K . For a one-particle probe $\langle\psi| = \int d^3p \psi^*(\mathbf{p})\langle p|$,

$$\langle\psi|a_1^\dagger(t)|0\rangle = (2\pi)^3 \int d^3p \psi^*(\mathbf{p}) 2\omega f_1(\mathbf{p}) \langle p|\varphi(0)|0\rangle, \quad (10.6.5)$$

independent of t . By Lorentz invariance $\langle p|\varphi(0)|0\rangle$ is a constant (LecNote4 p. 7). The one-particle part of $a_1^\dagger(t)|0\rangle$ is this constant times the free-theory packet, at every t , and the limit cannot change it. Its normalization is fixed by the rescaling $\langle p|\varphi(0)|0\rangle = 1$ (LecNote4 p. 8).

Two or more particles. A state $|p, n\rangle$ with total momentum $\mathbf{p}$ has $E = \sqrt{\mathbf{p}^2 + M^2}$, with invariant mass $M \geq 2m$ running over a continuum. Then $K = \sqrt{\mathbf{p}^2 + M^2} - \sqrt{\mathbf{p}^2 + m^2} > 0$ varies with $\mathbf{p}$ and with M , and (5.23) goes to zero. No condition on φ is needed. The same holds for a bound state of mass $M \neq m$: it is an isolated point of the mass spectrum, but its K still varies with $\mathbf{p}$.

The physical picture is the one on LecNote4 p. 10. The operator $a_1^\dagger(t)$ builds its packet with the one-particle dispersion $\omega(\mathbf{k})$. The part of $a_1^\dagger(t)|0\rangle$ that really is one particle moves with that dispersion and stays with the packet. The multi-particle part moves with group velocity $\mathbf{p}/\sqrt{\mathbf{p}^2 + M^2} \neq \mathbf{p}/\omega$ and drifts away from any fixed state. The vacuum cannot drift; it is everywhere.

Annihilation operator. The same algebra with $a(\mathbf{k}, t) = i \int d^3x e^{-ikx} \overleftrightarrow{\partial}_0 \varphi(x)$ gives

$$\langle s|a_1(t)|0\rangle = (2\pi)^3 [\omega(\mathbf{p}_s) - E_s] f_1^*(-\mathbf{p}_s) \langle s|\varphi(0)|0\rangle e^{i(\omega(\mathbf{p}_s) + E_s)t}. \quad (10.6.6)$$

1056 The one-particle part vanishes identically, at every t , because $E_s = \omega$. The
 1057 vacuum part is $\propto v e^{imt}$ and needs $v = 0$. The multi-particle part dephases.
 1058 Putting the sectors together, with $v = 0$ and $\langle p | \varphi(0) | 0 \rangle = 1$,

$$a_1^\dagger(t)|0\rangle \rightarrow \int d^3k f_1(\mathbf{k}) |k\rangle, \quad a_1(t)|0\rangle \rightarrow 0 \quad (t \rightarrow \pm\infty, \text{ weakly}). \quad (10.6.7)$$

1059 These are the properties the derivation in LecNote4 uses. The limit is
 1060 weak: the multi-particle part of $a_1^\dagger(t)|0\rangle$ keeps its norm, it just stops
 1061 overlapping with anything fixed. For one operator acting on the vacuum
 1062 this is all LSZ needs. Making it rigorous for products of several $a^\dagger(-\infty)$ is
 1063 the content of Haag–Ruelle scattering theory. (Strictly, in an interacting
 1064 theory in four dimensions a field smeared in space at a single time is too
 1065 singular for the norm of $a_1^\dagger(t)|0\rangle$ to be finite; Haag–Ruelle theory smears in
 1066 time as well. The conclusions carry over.)

1067 HOW FAST THE MULTI-PARTICLE PART GOES AWAY

1068 To see the rates, take the probe equal to the packet and treat the smooth
 1069 prefactor in (10.6.2) as constant across it ($\sigma \ll m$). Normalized to $A(0) = 1$,

$$A(t) = \int d^3p w(\mathbf{p}) e^{iK(\mathbf{p})t}, \quad w(\mathbf{p}) = (2\pi\sigma^2)^{-3/2} e^{-(\mathbf{p}-\mathbf{k}_1)^2/2\sigma^2}, \quad (10.6.8)$$

1070 with $K(\mathbf{p}) = E_s(\mathbf{p}) - \omega(\mathbf{p})$ for the sector in question; $w = |f_1|^2$ up to
 1071 normalization. For a fixed-mass sector, $E_s = \sqrt{\mathbf{p}^2 + M^2}$.
 1072 *Gaussian estimate.* Expand K about the packet centre. The size of its
 1073 gradient there is the difference of group velocities,

$$\Delta v = \frac{k_1}{\omega(k_1)} - \frac{k_1}{\sqrt{k_1^2 + M^2}}, \quad (10.6.9)$$

1074 and keeping only the linear term turns (10.6.8) into the Fourier
 1075 transform of a Gaussian:

$$|A(t)| \approx e^{-(\sigma \Delta v t)^2/2}, \quad \tau = \frac{1}{\sigma \Delta v}. \quad (10.6.10)$$

1076 The two components separate at relative speed Δv and stop overlapping
 1077 once the separation exceeds the packet size, about $1/\sigma$. The two
 1078 directions transverse to $\mathbf{k}_1$ add a factor $[1 + ((1/\omega - 1/E_s)\sigma^2 t)^2]^{-1/2}$, which
 1079 only speeds up the decay. The cubic term in the expansion corrects
 1080 (10.6.10) once $|A|$ is down to a few per cent.

1081 *Power-law tail.* The gradient of K vanishes at $\mathbf{p} = 0$, where both
 1082 components are at rest and nothing moves apart. Near that point
 1083 $K \approx (M - m) - \frac{1}{2}(1/m - 1/M) \mathbf{p}^2$. For a packet at rest ($\mathbf{k}_1 = 0$) the Gaussian
 1084 integral is exact at this order:

$$A(t) = e^{i(M-m)t} [1 + i(1/m - 1/M) \sigma^2 t]^{-3/2}, \quad |A| \rightarrow [(1/m - 1/M) \sigma^2 t]^{-3/2}. \quad (10.6.11)$$

1085 For a moving packet, (10.6.10) dies first. The small part of the packet
 1086 near $p = 0$, weighted by $w(0) \propto e^{-k_1^2/2\sigma^2}$, then takes over and falls only as a
 1087 power of t . In a one-dimensional model along k_1 the stationary-phase
 1088 estimate of this tail is

$$|A_{\text{tail}}| \approx w(0) \left[\frac{2\pi}{(1/m - 1/M)t} \right]^{1/2}. \quad (10.6.12)$$

1089 *Continuum in M .* At fixed p the multi-particle sector also has a
 1090 continuum of M , so it dephases further. Near threshold the two-particle
 1091 weight typically grows like $(M - 2m)^{1/2}$, which gives another $t^{-3/2}$.
 1092 *Wrong mass.* If the m used to build $a^\dagger(k)$ is not the pole mass m' , the
 1093 one-particle $K = \sqrt{p^2 + m'^2} - \sqrt{p^2 + m^2}$ varies with p . The one-particle
 1094 overlap then decays too, following (10.6.10) with $M \rightarrow m'$, and the LSZ
 1095 formula returns zero. This is the dynamical reason the m in LSZ must be
 1096 the physical mass (LecNote4 p. 10).
 1097 *Numbers.* The table on p. 43 and Fig. 10.1 evaluate (10.6.8) in the
 1098 one-dimensional model along k_1 , with $m = 1$, $k_1 = m$ and $\sigma = 0.25m$. For
 1099 the shell at $M = 2m$, $\Delta v = 1/\sqrt{2} - 1/\sqrt{5} = 0.260$ and $\tau = 15.4$; for a packet
 1100 mass 5% off ($m' = 1.05m$), $\Delta v = 0.0175$ and $\tau = 229$. The Gaussian estimate
 1101 describes the 5% case throughout. For $M = 2m$ it holds to $t \approx 30$; the
 1102 cubic term $K'''(k_1) = 0.316$ accounts for the gap at $t = 40$ (the quadratic
 1103 term, $K'' = 0.004$, is negligible). From $t \approx 60$ on the power-law tail takes
 1104 over. Eq. (10.6.12), with $w(0) = 5.4 \times 10^{-4}$, is its asymptotic form: it is
 1105 within 30% of the exact value at $t = 160$ and within 10% for $t \gtrsim 300$,
 1106 because w varies quickly across the region near $p = 0$. In three
 1107 dimensions with the packet at rest, the numerical $|A|t^{3/2}$ approaches 180.9
 1108 at $t = 800$, against 181.0 from (10.6.11).

mt	packet mass 5% off		shell at $M = 2m$			
	exact	Eq. (10.6.10)	exact	Eq. (10.6.10)	cubic K	Eq. (10.6.12)
20	0.996	0.996	0.444	0.430	0.443	—
40	0.986	0.985	2.0×10^{-2}	3.4×10^{-2}	2.1×10^{-2}	—
80	0.944	0.941	4.8×10^{-4}	1.4×10^{-6}	—	2.1×10^{-4}
160	0.794	0.784	1.9×10^{-4}	3×10^{-24}	—	1.5×10^{-4}

$|A(t)|$ IN THE ONE-DIMENSIONAL MODEL

From (10.6.8) in one dimension, $m = 1$, $k_1 = m$, $\sigma = 0.25m$. The vacuum ($|A| = |e^{-imt}|$) and the one-particle sector with the correct mass ($K \equiv 0$) are exactly 1 at all t . The cubic expansion of K is shown only while it applies; at late times it has spurious stationary points far out in the packet.

1109 THE IN AND OUT VACUUM

1110 On LecNote4 p. 5, after time ordering, the operators $a_{i'}(-\infty)$ act on the
 1111 ket and the $a_j^\dagger(+\infty)$ act on the bra. So the ket must satisfy $a(-\infty)|0\rangle = 0$:

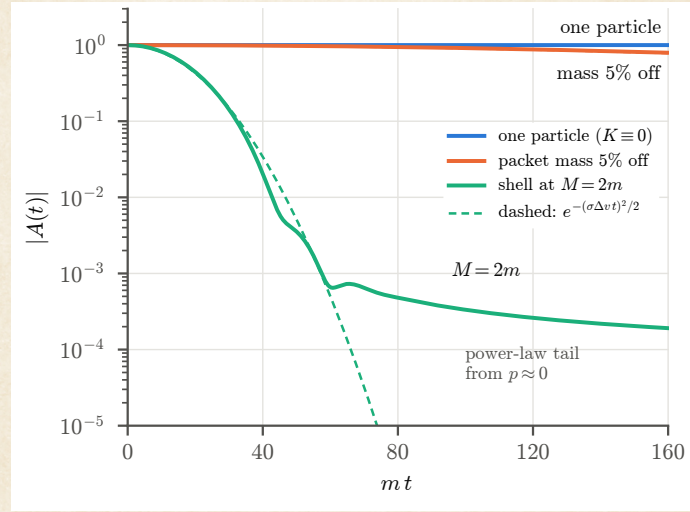


Figure 10.1: $|A(t)|$ for the one-particle sector ($K \equiv 0$), a packet built with a mass 5% off, and a shell at $M = 2m$; parameters as in the table on p. 43. Dashed: the Gaussian estimate (10.6.10) (for the 5% case it lies on the solid curve).

1112 it is the in-vacuum. The bra must satisfy $\langle 0|a^\dagger(+\infty) = 0$, i.e. $a(+\infty)|0\rangle = 0$:
 1113 it is the out-vacuum. In the Heisenberg picture states carry no time label;
 1114 “in” and “out” say which asymptotic operators annihilate the state. By
 1115 (10.6.7), the exact ground state, with $v = 0$, is annihilated at both ends.
 1116 So $|0_{\text{in}}\rangle = |0_{\text{out}}\rangle = |0\rangle$, and

$$\langle 0|0\rangle = 1 \quad \Longleftrightarrow \quad \langle 0_{\text{out}}|0_{\text{in}}\rangle = 1. \quad (10.6.13)$$

1117 The content is that the vacuum is stable: no particles are produced from
 1118 it. That needs a unique ground state and a time-independent H . In the
 1119 Heisenberg picture, with one vacuum state, no phase can appear. In the
 1120 path integral the same amplitude is computed as $\langle 0|e^{-iHT}|0\rangle = e^{-iE_0T}$, and
 1121 there it needs the choice $E_0 = 0$ (LecNote5 p. 8 and §10.8).

1122 LecNote4 p. 7 uses that both vacua are annihilated by P^μ . The step
 1123 $\langle 0|\varphi(x)|0\rangle = \langle 0|\varphi(0)|0\rangle$ uses $\langle 0|P^\mu = 0$ on the bra and $P^\mu|0\rangle = 0$ on the ket.
 1124 That is why v is a constant, and why $v = 0$ as $t \rightarrow \pm\infty$ is the same as $v = 0$
 1125 everywhere.

1126 A time-dependent external source shows what happens otherwise. Add
 1127 $J(x)\varphi(x)$ to the free Lagrangian, with J switched on and off. The
 1128 vacuum-to-vacuum amplitude is (Srednicki §8)

$$\langle 0_{\text{out}}|0_{\text{in}}\rangle_J = \exp\left[\frac{i}{2} \int d^4x d^4y J(x)\Delta(x-y)J(y)\right], \quad (10.6.14)$$

1129 and with $\text{Im}(k^2 + m^2 - i\epsilon)^{-1} = \pi\delta(k^2 + m^2)$ its squared modulus is

$$|\langle 0_{\text{out}}|0_{\text{in}}\rangle_J|^2 = \exp\left[-\int \widetilde{d\mathbf{k}} |\tilde{J}(\omega, \mathbf{k})|^2\right], \quad \tilde{J}(k) = \int d^4x e^{-ikx} J(x). \quad (10.6.15)$$

1130 This is less than 1 whenever J has Fourier components on the mass
 1131 shell: the source creates particles, on average $\int \widetilde{dk} |\tilde{J}|^2$ of them. The
 1132 in-out expectation value of the field,

$$\frac{\langle 0_{\text{out}} | \varphi(x) | 0_{\text{in}} \rangle_J}{\langle 0_{\text{out}} | 0_{\text{in}} \rangle_J} = \int d^4 y \Delta(x - y) J(y), \quad (10.6.16)$$

1133 depends on x : it solves $(-\partial^2 + m^2)\varphi = J$. With time translation broken, the
 1134 vacua are no longer eigenstates of P^0 , the in and out vacua differ, and a
 1135 condition “at $t \rightarrow \pm\infty$ ” is no longer the same as one “at all times”. For
 1136 $J = 0$ both statements reduce to (10.6.13).

1137 TIME ORDERING AND THE OVERLAP TERM

1138 After substituting (5.11) and (5.12), LecNote4 p. 5 uses T to move every
 1139 $a^\dagger(+\infty)$ to the far left and every $a(-\infty)$ to the far right, where they vanish
 1140 by (10.6.7). What survives is the LSZ integral. It reproduces $\langle f|i \rangle$ up to
 1141 terms of order $e^{-(\mathbf{k}_{i'} - \mathbf{k}_j)^2/8\sigma^2}$: the step drops the part of $\langle f|i \rangle$ in which
 1142 nothing scatters. Write the overlap of two of the one-particle packets in
 1143 (10.6.7) as

$$O_{i'j} \equiv (2\pi)^3 \int d^3 k \, 2\omega f_{i'}^*(\mathbf{k}) f_j(\mathbf{k}) \approx e^{-(\mathbf{k}_{i'} - \mathbf{k}_j)^2/8\sigma^2}, \quad (10.6.17)$$

1144 where the Gaussian form follows from completing the square, with ω
 1145 roughly constant across the packets. For $2 \rightarrow 2$ with stable particles,

$$\langle f|i \rangle = O_{1'1} O_{2'2} + O_{1'2} O_{2'1} + (\text{LSZ formula}). \quad (10.6.18)$$

1146 The overlap terms are the part of $S = 1 + iT$ in which nothing scatters.
 1147 Choosing every $|\mathbf{k}_{i'} - \mathbf{k}_j| \gg \sigma$, as on LecNote4 p. 3, makes them
 1148 exponentially small. In the plane-wave limit they become the $\delta^3(\mathbf{k}_{i'} - \mathbf{k}_j)$
 1149 terms of the “1”.

1150 *Free-theory check.* In the free theory $(-\partial^2 + m^2)\varphi = 0$ as an operator, so
 1151 (5.11) and (5.12) say $a^\dagger(-\infty) = a^\dagger(+\infty)$ and $a(+\infty) = a(-\infty)$. The integrals
 1152 vanish, and the only term left after the substitution is
 1153 $\langle 0|T a_{1'}(-\infty) a_{2'}(-\infty) a_2^\dagger(+\infty) a_1^\dagger(+\infty)|0 \rangle$, which the step on p. 5 sets to zero.
 1154 But these are the same four operators as in the original product, whose
 1155 value is $O_{1'1} O_{2'2} + O_{1'2} O_{2'1}$. The labels $\pm\infty$ do not order them in any
 1156 physical sense. The step drops exactly the no-scattering part, as (10.6.18)
 1157 says.

1158 In the interacting theory the dropped pieces come from the ends of the
 1159 time integrals in (5.11) and (5.12), which run to the same asymptotic
 1160 times as the operators they are ordered against. Individually these pieces
 1161 also contain multi-particle terms; their total is the overlap terms in
 1162 (10.6.18), which we do not show here. Every one of them contains

1163 $f_i^*(\mathbf{p})f_j(\mathbf{p})$ at a common momentum, so all of them vanish when every
 1164 $|\mathbf{k}_i - \mathbf{k}_j| \gg \sigma$.

A two-mode model. Take two oscillators of the same frequency with a coupling switched on and off, $H = \omega(a^\dagger a + b^\dagger b) + g(t)(a^\dagger b + b^\dagger a)$. In the rotating frame $\tilde{a}(t) = e^{i\omega t} a(t)$ the free evolution drops out, which is the job $(-\partial^2 + m^2)$ does in LSZ, and $\dot{\tilde{a}} = -ig\tilde{b}$, $\dot{\tilde{b}} = -ig\tilde{a}$. With $\theta(t) = \int_{-\infty}^t g dt'$ and $\Theta = \theta(\infty)$, $\tilde{a}(t) = \cos \theta \tilde{a}_{\text{in}} - i \sin \theta \tilde{b}_{\text{in}}$. The analogue of (5.12) is $\tilde{a}_{\text{out}} = \tilde{a}_{\text{in}} - i \int dt g(t) \tilde{b}(t)$, and

$$\langle 0 | \tilde{a}_{\text{out}} \tilde{a}_{\text{in}}^\dagger | 0 \rangle = \underbrace{\langle 0 | \tilde{a}_{\text{in}} \tilde{a}_{\text{in}}^\dagger | 0 \rangle}_1 - \int dt g \sin \theta = \cos \Theta, \quad (10.6.19)$$

$$\langle 0 | \tilde{b}_{\text{out}} \tilde{a}_{\text{in}}^\dagger | 0 \rangle = \underbrace{\langle 0 | \tilde{b}_{\text{in}} \tilde{a}_{\text{in}}^\dagger | 0 \rangle}_0 - i \int dt g \cos \theta = -i \sin \Theta. \quad (10.6.20)$$

1165 The integral, the analogue of the LSZ integrand, gives the full amplitude
 1166 when the initial and final states differ, and the amplitude minus the
 1167 overlap, $\cos \Theta - 1$, when they coincide. Moving $\tilde{a}_{\text{in}}$ to the right, where it
 1168 annihilates $|0\rangle$, is exactly the step that drops the underbraced 1.

Derivatives outside T . The factors $(-\partial^2 + m^2)$ in the LSZ formula act outside the time ordering. Moving ∂_0^2 inside produces contact terms from the equal-time commutator,

$$\begin{aligned} & (-\partial_x^2 + m^2) \langle 0 | T \varphi(x) \varphi(y) X | 0 \rangle \\ &= \langle 0 | T [(-\partial^2 + m^2) \varphi(x)] \varphi(y) X | 0 \rangle - \frac{i}{Z_\varphi} \delta^4(x - y) \langle 0 | T X | 0 \rangle + \dots, \end{aligned} \quad (10.6.21)$$

1169 where X is the product of the remaining fields and the dots are the same
 1170 contact term for each field in X . The factor Z_φ is the one on LecNote4
 1171 p. 10: for the field normalized by $\langle p | \varphi(0) | 0 \rangle = 1$ the canonical momentum
 1172 is $Z_\varphi \partial_0 \varphi$ (in the free theory $Z_\varphi = 1$). A contact term sets two positions
 1173 equal, so its Fourier transform depends on the two external momenta
 1174 only through one combination. It cannot have a separate pole in each of
 1175 the two k^2 , and for non-forward kinematics, $|\mathbf{k}_i - \mathbf{k}_j| \gg \sigma$, the factors
 1176 $k^2 + m^2 \rightarrow 0$ remove it on shell.